PROJECT W.A.F.F.L.E.
OFFLINE FIELD CONSOLE // MAP STACK

ZERO-SERVER ARCHITECTURE

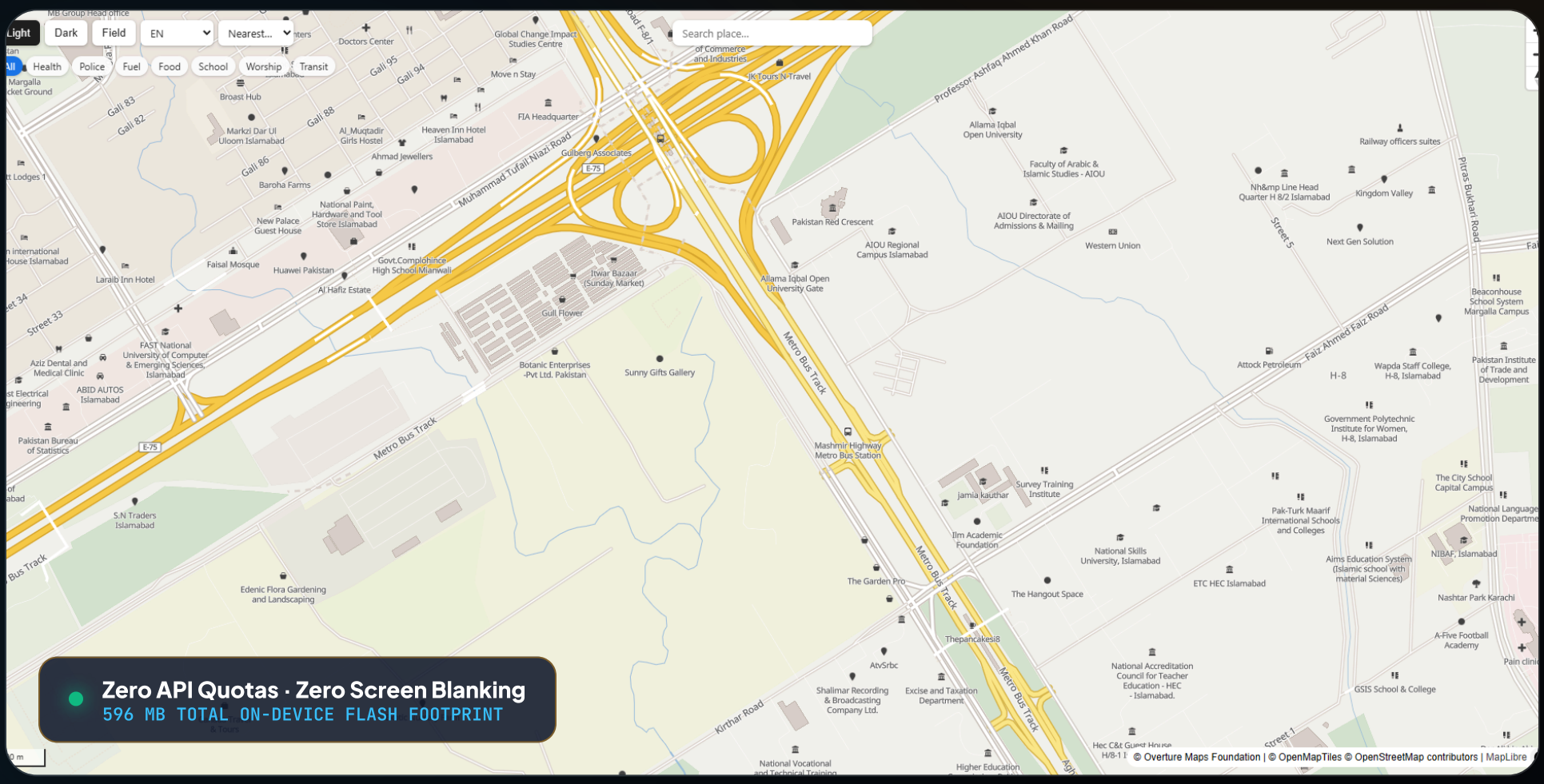
A 100% Offline Vector Map of Pakistan.

Replaced cloud tile dependency with 253,618 vector tiles, 809k Overture amenities, bilingual Urdu labels, and instant on-device FTS5 search.

253.6k
Vector Tiles (z0–14)
OpenMapTiles MVT gzip

809,824
Overture Places
Layer: place · phone · addr

< 5 ms
Offline Search
SQLite FTS5 Prefix + BM25

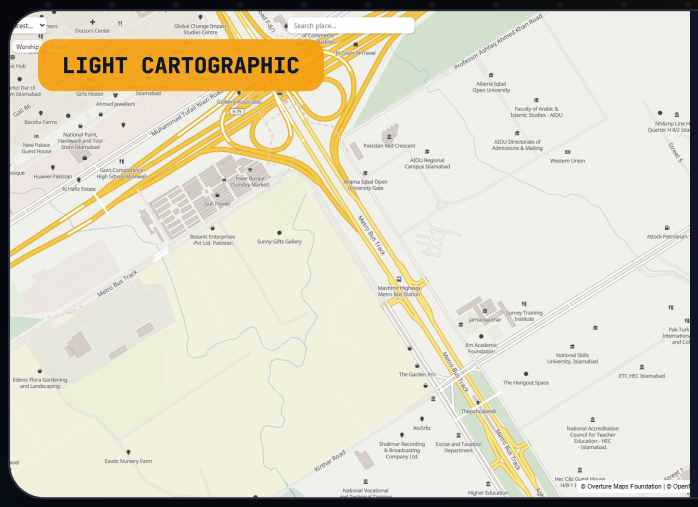


github.com/AhmadHasssan9181/waffle-maps

SWIPE ➜

Field-Tuned Visual Ergonomics.

Every style is authored in 3 script variants: **English**, **Urdu (name:ur)**, and **Stacked Bilingual** — backed by all 256 Unicode glyph ranges so Arabic/Urdu never drops a block.



Daylight Arterial Hierarchy

High-readability golden highway conduits, soft terrain fills, 3D building extrusions, and categorized POI sprite overlays.

`style_light.json` `EN / UR / BOTH` `Day Touring`



🌙 Low-Glare Cockpit Slate

Deep charcoal canvas engineered for zero dashboard reflection and rapid night navigation across rural corridors.

`style_dark.json` `47 Layers` `Night Operations`



🚨 Tactical High-Luminance Amber

Pure black substrate paired with high-contrast amber vectors for direct desert/mountain sunlight and emergency HUD monitors.

`style_field.json` `OLED Power Saver` `Sunlight HUD`

809,824 Offline Places at Your Fingertips.

Google raster tiles are just flat pictures — they can't be searched or tapped. We extracted Overture Maps places via DuckDB `ST_AsMVT` to build a real interactive spatial graph.

VECTOR LAYER — 214.5 MB

809,824

Interactive Map Points

Clustered in `pakistan_places.pmtiles` (z11–14). Tapping any point returns name, category, phone number, address, and status.

INDEX ENGINE — 104.4 MB

< 5 ms

SQLite FTS5 Full-Text Index

Zero-server prefix search & BM25 ranking across all national POIs. Finds hospitals, fuel stations, banks, and bazaars instantly.

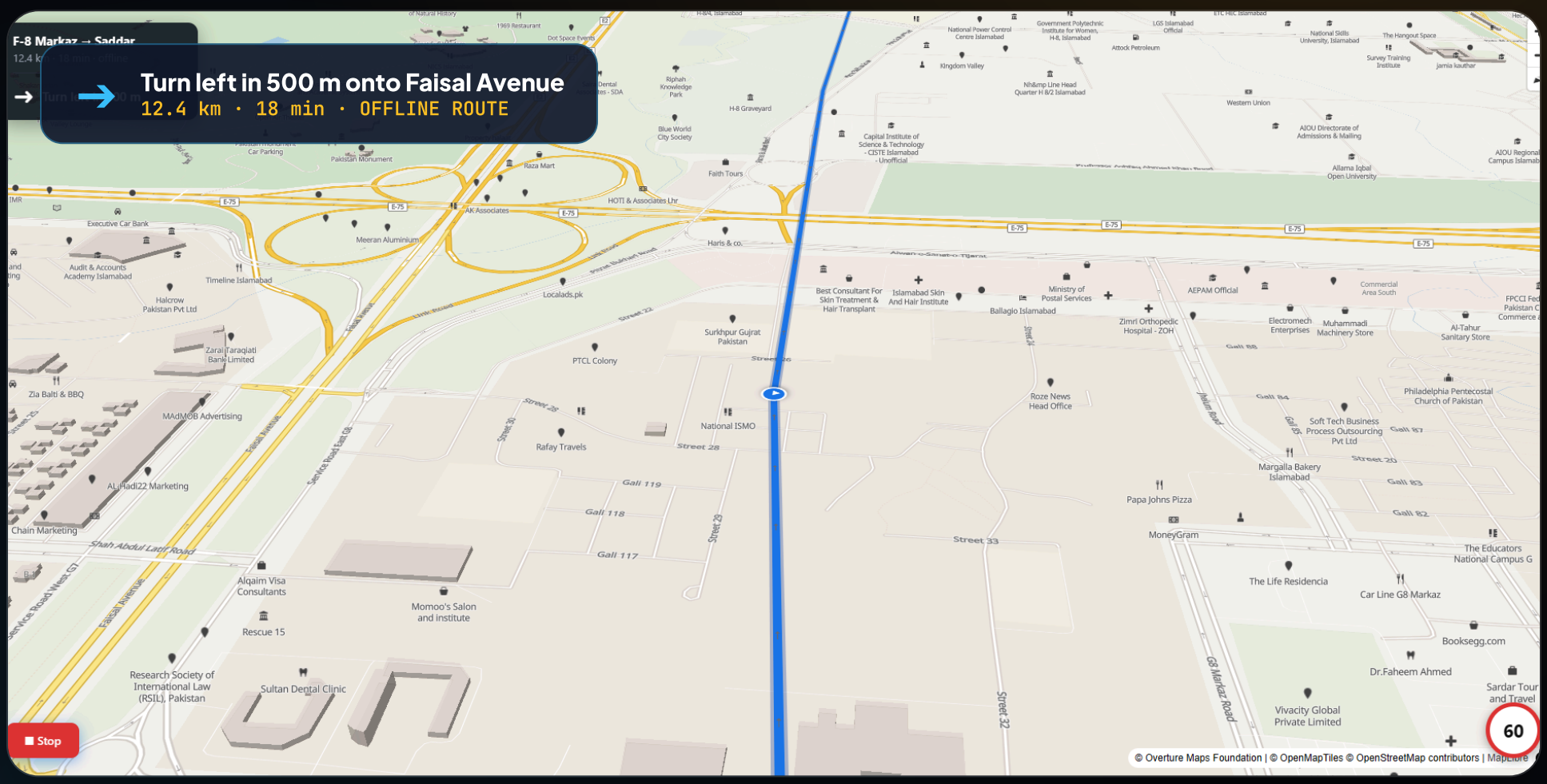
places_search.db — Prefix + BM25 Query

```
SELECT p.name, p.cat, p.lon, p.lat
FROM places_fts f JOIN places p ON p.id = f.rowid
WHERE places_fts MATCH 'hospital*'
ORDER BY bm25(places_fts) LIMIT 20; -- executes in 3.8ms
```

🏥 Medical & Hospitals (14.2k) · ⛽ Fuel & EV (8.4k) · 🕌 Places of Worship (92k) · 🏫 Education (38k) · 🍔 Food & Grocery (110k)

Navigation-Ready Vector Rendering.

Vector maps aren't just for looking at—they power 60 FPS turn-by-turn guidance with dynamic camera tilt, multi-layer route casing, and offline GraphHopper routing.



60° Camera Pitch

Smooth prospective viewing angle for situational anticipation.

Dual-Layer Casing

Route casing + inner blue line with directional pulse arrows.

Speed Limit HUD

Road-classified legal speed limit indicator with overspeed alert.

Built for Zero-Signal Sovereignty.

The complete pipeline, compilation scripts, styles, and download links are public and open-source.

01	**OpenMapTiles Basemap PMTiles** Planetiler 0.10.2 · Geofabrik OSM PBF · z0–z14	266.1 MB
02	**Overture Maps Places PMTiles** DuckDB Spatial ST_AsMVT · 809,824 POIs · z11–14	214.5 MB
03	**SQLite FTS5 Search Index** unicode61 tokenizer · prefix matching & BM25 rank	104.4 MB
04	**9 Styles & All 256 Glyph Ranges** Noto Sans Regular/Medium/Italic · 1×/@2× sprites	11.0 MB

Download the Data Pack & Build Scripts

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